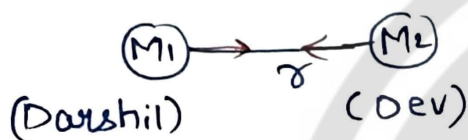


GRAVITATION

Introduction:

Gravitation is a natural phenomenon by which:

- Every object in the universe attracts every other object.
- This attraction depends on their masses and the distance between them.
- It is a Universal force - acts everywhere, on all bodies, big or small.



Universal Law of Gravitation:

If two bodies of masses m_1 and m_2 are separated by a distance r , then the force of attraction is described by the following relationship.

$$F = G \frac{m_1 m_2}{r^2}$$

$F \propto m_1$
 $F \propto m_2$
 $F \propto \frac{1}{r^2}$

Where,

F = Gravitational force between the two bodies.

G = Universal Gravitational constant.

m_1, m_2 = masses of the two bodies
 r = dist b/w the centres of the two bodies.

Acceleration due to Gravity (g):

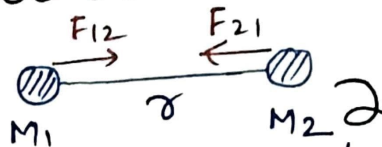
- Acceleration
- $g = 9.8 \text{ m/s}^2$
 $\approx 10 \text{ m/s}^2$

Universal Gravitation constant (G)

- It is constant
- $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

The Gravitational constant:

$$F = \frac{G m_1 m_2}{r^2}$$



G = Universal Gravitational constant

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$F = \frac{G m_1 m_2}{r^2} \quad \text{Units: } \frac{\text{kg} \cdot \text{kg}}{\text{m}^2} \rightarrow \text{m}^2 \therefore G = \text{Nm}^2/\text{kg}^2$$

$F_{12} = -F_{21} = F$

Dimensional Formula of G :

$$F = \frac{G m_1 m_2}{r^2} \quad \Rightarrow G \rightarrow \text{unit} = \frac{\text{kg m s}^{-2} \cdot \text{m}^2}{\text{kg}^2}$$

$$\Rightarrow G = \frac{F r^2}{m_1 m_2}$$

$$\Rightarrow \frac{\text{m}^3 \text{s}^{-2}}{\text{kg}} = \text{kg}^{-1} \text{m}^3 \text{s}^{-2}$$

Dimensional Formula:

$$G = [M^{-1} L^3 T^{-2}]$$

Acceleration due to Gravity:



Below depth 'd' from the Earth's surface

on the surface of Earth

Above the height 'h' from the Earth's surface.

Acceleration due to Gravity of the Earth:

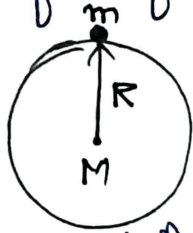
Gravitational force between the Earth (Mass M) and the object (of mass m) on the surface of Earth.

$$F = \frac{G M m}{R^2}$$

Since, we know that, $F = ma$

$$\Rightarrow ma = \frac{GMm}{R^2}$$

$\Rightarrow$ Acceleration due to gravity on the surface of earth



$$g = \frac{GM}{R^2}$$

$$\Rightarrow g = 9.8 \text{ m/s}^2 \approx 10 \text{ m/s}^2$$

R = Radius of Earth = 6400 km.

Only valid for the case when we calculate the grav. force on the surface of Earth.

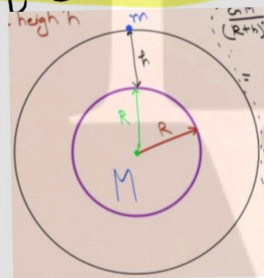
$$F = \frac{GMm}{R^2}$$

Puri calculation nhi karne ka

Final answer $F = mg$ se nikalne ka.

Acceleration due to Gravity - Above the Surface of the Earth:

Let's say that the accn due to gravity above 'h' from the surface of earth is 'g'



$$F = \frac{GMm}{(R+h)^2} \Rightarrow mg' = \frac{GMm}{(R+h)^2}$$

$$\Rightarrow g' = \frac{GM}{(R+h)^2} \Rightarrow g' = \frac{GM}{R^2 \left(1 + \frac{h}{R}\right)^2}$$

$$\Rightarrow g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2} \Rightarrow g' = g \left(\frac{R}{R+h}\right)^2$$

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2} \Rightarrow g' = g \left(1 + \frac{h}{R}\right)^{-2}$$

Now if the height 'h' is not very far from the earth's surface, then $\frac{h}{R} \ll 1$.

$\therefore$ By Binomial $(1+x)^{-n} = 1 - nx + \dots$
Expansion $\Rightarrow (1+x)^{-n} = (1 - nx)$

Where $x \ll 1$

$$\Rightarrow g' = g \left(1 - \frac{2h}{R}\right)$$

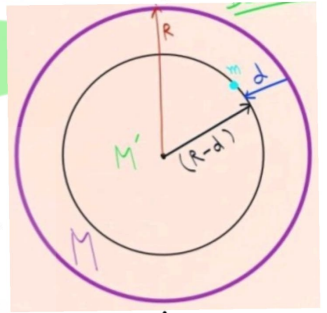
$\rightarrow$ Only valid when $\frac{h}{R} \ll 1$
or $h \ll R$.

Acceleration due to gravity below the Surface of Earth. v. Imp Derivation

Density of Earth = $\frac{\text{Mass of Earth}}{\text{Volume of Earth}}$

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} \text{ --- (1)}$$

Let the Mass of Sphere with radius (R-d) be M' :



$$\rho = \frac{M'}{\frac{4}{3}\pi (R-d)^3} \Rightarrow \frac{M}{\frac{4}{3}\pi R^3} = \frac{M'}{\frac{4}{3}\pi (R-d)^3}$$

$$\Rightarrow M' = \frac{M(R-d)^3}{R^3} \Rightarrow M' = M \left(\frac{R-d}{R}\right)^3$$

$$\Rightarrow M' = M \left(1 - \frac{d}{R}\right)^3 \text{ --- (2)}$$

Now the force of gravitation on mass is:

$$F = \frac{GM'm}{(R-d)^2} \Rightarrow mg' = \frac{GM'm}{(R-d)^2}$$

$$\Rightarrow g' = \frac{G}{(R-d)^2} \cdot M \left(1 - \frac{d}{R}\right)^3$$

$$\Rightarrow g' = \frac{GM}{R^2 \left(1 - \frac{d}{R}\right)^2} \times \left(1 - \frac{d}{R}\right)^3$$

$\Rightarrow g' = g \left(1 - \frac{d}{R}\right)$ $\rightarrow$ Accn due to gravity becoz of earth on an object placed at depth 'd' below the earth's surface.

Escape Speed:

The Minimum velocity with which a body must be projected from the surface of a planet (or any celestial body) so that it escapes from the planet's gravitational field without ever returning is called the escape speed.

$$V_e = \sqrt{2gR} \Rightarrow V_e = \sqrt{2 \frac{GM}{R^2} \times R}$$

$\rightarrow g = \text{Accn}$
due to gravity
of the Planet

$\rightarrow R = \text{Radius}$
of that Planet

$$V_e = \sqrt{\frac{GM \times 2}{R}} = \sqrt{\frac{2GM}{R}}$$

$\rightarrow G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
 $\rightarrow M = \text{Mass of the Planet}$
 $\rightarrow R = \text{Radius of the Planet}$

Gravitational Potential Energy:

Gravitational Potential Energy (GPE) is the energy possessed by a body due to its position in the gravitational field of the Earth (or any other massive body).

When a body of mass 'm' is at a distance 'r' from the centre of the Earth of Mass M, then its GPE is:

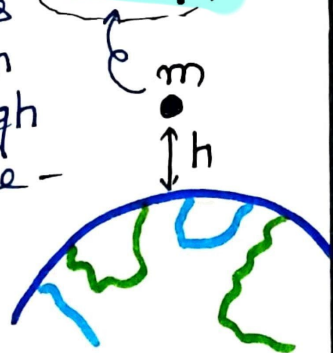
When the height 'h' of an object of mass m is too high, then we can't use $U = mgh$ instead we will use -

$$U = -\frac{GMm}{r}$$

$\rightarrow \text{Scalar Quantity}$

$\rightarrow \text{S.I Unit} \rightarrow \text{Joule (J)}$

$$P.E = mgh \Rightarrow U$$



DERIVATION:

It is the work done in bringing the body from infinity to a particular point in the gravitational field.

$$U = W = \int_{\infty}^r F \cdot dx$$

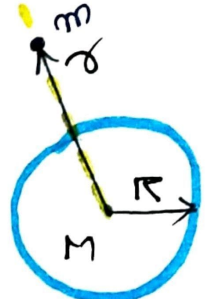
$$\Rightarrow U = \int_{\infty}^r \frac{GMm}{x^2} \cdot dx$$

$$\Rightarrow U = GMm \int_{\infty}^r (x^{-2}) dx$$

$$\Rightarrow U = GMm \left[-\frac{1}{x} \right]_{\infty}^r$$

$$\Rightarrow U = GMm \left[\left(-\frac{1}{r} \right) - \left(-\frac{1}{\infty} \right) \right] \left\{ \begin{array}{l} \frac{x^{-2+1}}{-2+1} \\ = \frac{x^{-1}}{-1} \\ = \left(-\frac{1}{x} \right) \end{array} \right.$$

$$U = -\frac{GMm}{r}$$



Gravitational Potential: (V)

Gravitational Potential at a point in a gravitational field is the work done in bringing a unit mass (mass = 1 kg) from infinity to that point. It is a Scalar quantity. $\rightarrow \text{No direction}$

Unit charge $q = 1 \text{ C}$

$\rightarrow \text{one} \rightarrow \text{P.E} \rightarrow \text{①}$

$$V = W = \int_{\infty}^r F \cdot dx \quad \left\{ F = \frac{GM(1)}{x^2} \right.$$

$$\Rightarrow V = \int_{\infty}^r \frac{GM}{x^2} \cdot dx$$

$$\Rightarrow V = GM \int_{\infty}^r (x^{-2}) dx$$

$$\Rightarrow V = GM \left[-\frac{1}{x} \right]_{\infty}^r$$

$$\Rightarrow V = -\frac{GM}{r}$$

$\rightarrow \text{S.I Unit}:$
Joule/kilogram.



Difference Between Potential and Potential Energy:

Gravitational Potential (v):
Work done per unit mass in bringing it from infinity to a Point

Gravitational Potential Energy (U):
Work done in bringing the entire mass (m) from infinity to a Point.

$$U = -\frac{GMm}{r} \quad U = mv^2 \quad v = -\frac{GM}{r}$$

Relation between G.P.E and G.P

Escape Speed: Value of escape speed on Earth =
(V_e)_{Earth} = 11.2 km/s

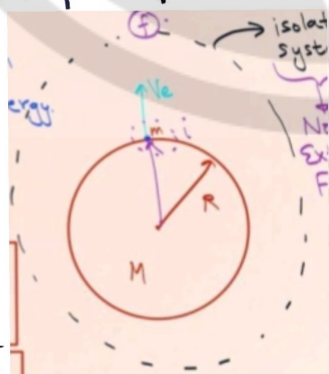
The Minimum velocity with which a body must be projected from the surface of a planet (or any celestial body) so that it escapes from the planet's gravitational field without ever returning is called the escape speed.

Since, no external force is acting on the system of a body of mass m and the planet of mass ' M ' therefore we can conserve the total energy.

$$E_i = E_f$$

$$\Rightarrow \frac{1}{2}mv_e^2 + \left(-\frac{GMm}{R}\right) = 0 + 0$$

$$\Rightarrow \frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0$$



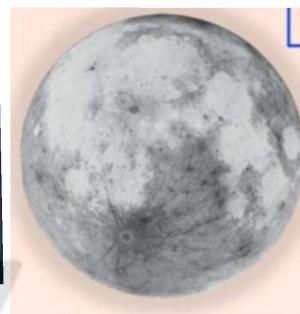
$$\Rightarrow \frac{v_e^2}{2} = \frac{GM}{R} \Rightarrow v_e^2 = \frac{2GM}{R}$$

$$\Rightarrow v_e = \sqrt{\frac{2GM}{R}}$$

$$v_e = \sqrt{2gR}$$

Earth Satellites:

- A satellite is a natural or artificial body revolving around a planet under the influence of its gravitational force.
- The Moon is a natural satellite of Earth.
- Man-made (artificial) satellites are launched into orbits around Earth for communication, weather, navigation, research etc.



Orbital Velocity:

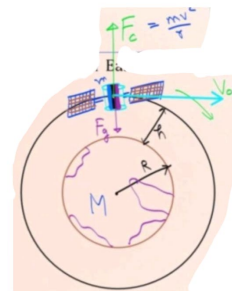
The minimum velocity required for a satellite to revolve around Earth in a stable circular orbit close to Earth's surface.

In order to revolve the satellite with orbital velocity v_o in the orbit around the Earth, we need to equalize the Gravitational Force and the Centripetal force.

$$F_c = F_g$$

$$\Rightarrow \frac{mv_o^2}{(R+h)} = \frac{GMm}{(R+h)^2}$$

$$v_o = \sqrt{\frac{GM}{R+h}}$$



Time Period of a Satellite:

$T = \frac{2\pi(R+h)}{V_0}$ satellite revolving very close to the earth $h=0$
 $T = 2\pi \sqrt{\frac{R^3}{GM}}$
 $\Rightarrow T = \frac{2\pi(R+h)}{(GM)^{1/2}}$
 $= \frac{2\pi(R+h)^{3/2}}{\sqrt{GM}}$
 $T^2 = \frac{2\pi(R+h)^3}{GM}$
 $T^2 \propto R^3$

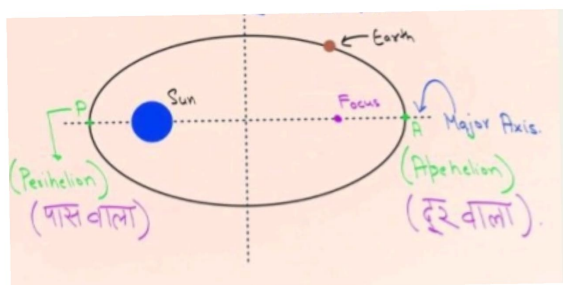
Kepler's Law:

- Johannes Kepler (1571-1630), a German astronomer, studied the motion of planets around the Sun.
- He analysed the astronomical data of Tycho Brahe and proposed three empirical laws of planetary motion, known as Kepler's Law.
- These laws laid the foundation for Newton's Law of Gravitation.



Kepler's First Law:

Law of Orbits:
 Each planet moves around the Sun in an elliptical orbit with the Sun at one of the two foci.



Perihelion:

End point of the major axis of the ellipse which is near to the Sun.

Aphelion:

End point of the major axis of the ellipse which is far from the Sun.

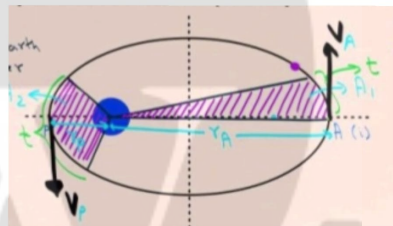
Kepler's Second Law:

Law of Areas:

The line joining a planet and the Sun sweeps out equal areas in equal intervals of time.

When at Perihelion, the Earth revolves or moves faster than when it is at Aphelion.

$$A_1 = A_2$$



Since, no external force or torque is being applied to the system of Sun and the Earth, thus we can conserve Angular Momentum.

$$L_A = L_P$$

$$\Rightarrow M V_A \delta A = M V_P \delta P$$

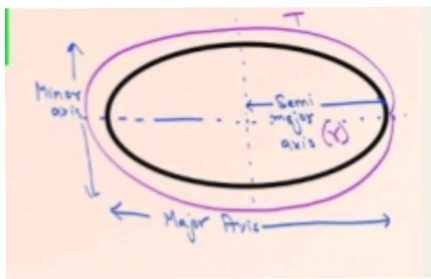
$$\frac{V_A}{V_P} = \frac{\delta P}{\delta A}$$

Kepler's Third Law:

Law of Periods:

The square of the time of revolution of a planet is proportional to the cube of the semi-major axis of the ellipse traced out by the planets.

$$T^2 \propto R^3$$



Energy of an Orbiting Satellite:

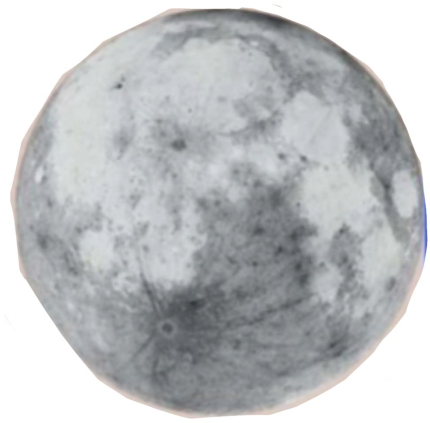
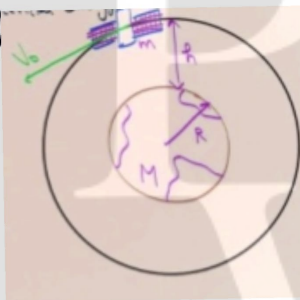
Total energy of a satellite revolving around the earth in a circular orbit will be the sum of its kinetic and Gravitational Potential Energy.

$$E = (K.E) + (P.E) \quad \text{where } v_0 = \sqrt{\frac{GM}{R+h}}$$

$$\Rightarrow E = \frac{1}{2} m (v_0)^2 + \left(-\frac{GMm}{(R+h)} \right)$$

$$\Rightarrow E = \frac{1}{2} m \frac{GM}{(R+h)} - \frac{GMm}{(R+h)}$$

$$\Rightarrow E = -\frac{GMm}{2(R+h)}$$



Some Important facts about Moon-

$$V_e = 11.2 \text{ km/s}$$

$$(V_e)_{\text{Moon}} = 2.38 \text{ km/s}$$

Escape velocity from the surface of Moon.

$$(Radius)_{\text{Moon}} = \left(\frac{1}{4} \right) (Radius)_{\text{Earth}}$$

Accn due to gravity on Moon:

$$(g)_{\text{Moon}} = \left(\frac{1}{6} \right) (g)_{\text{Earth}}$$